

Time Series Forecasting: An Exploratory Study Using S&P 500 ETF Data

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Abstract

Background

This study explores time series forecasting methods applied to the SPY-USA ETF using ~10 years of monthly price data (2016–2025). Financial markets are volatile and difficult to predict, making forecasting both valuable and challenging. Several models were tested, including naïve, seasonal naïve, drift, exponential smoothing (ETS), and an ensemble approach. Results show that simple trend-based models outperform seasonal approaches, with the additive ETS model providing the most accurate forecasts.

Methodology

Monthly price data spanning 10 years (2016 to 2025) for the SPY-USA was analyzed from FactSet. Exploratory analysis included trend visualization, autocorrelation, seasonal-trend using LOESS (STL) decomposition, and inflation-adjusted comparisons. The following models were then tested to forecast the SPY-USA ETF price: ETS, drift, naïve, seasonal naïve, and ensemble.

Results

The additive ETS model (Holt's Linear Trend) performed best, with the drift model performing similarly well, capturing the upward trend. The basic naïve model provided a reasonable baseline but lacked trend modeling. The seasonal naïve approach under-predicted the strong upward trend and exhibited autocorrelated residuals, making it the least reliable. The ensemble model performance was reduced due to inclusion of the weaker seasonal model.

Introduction & Significance

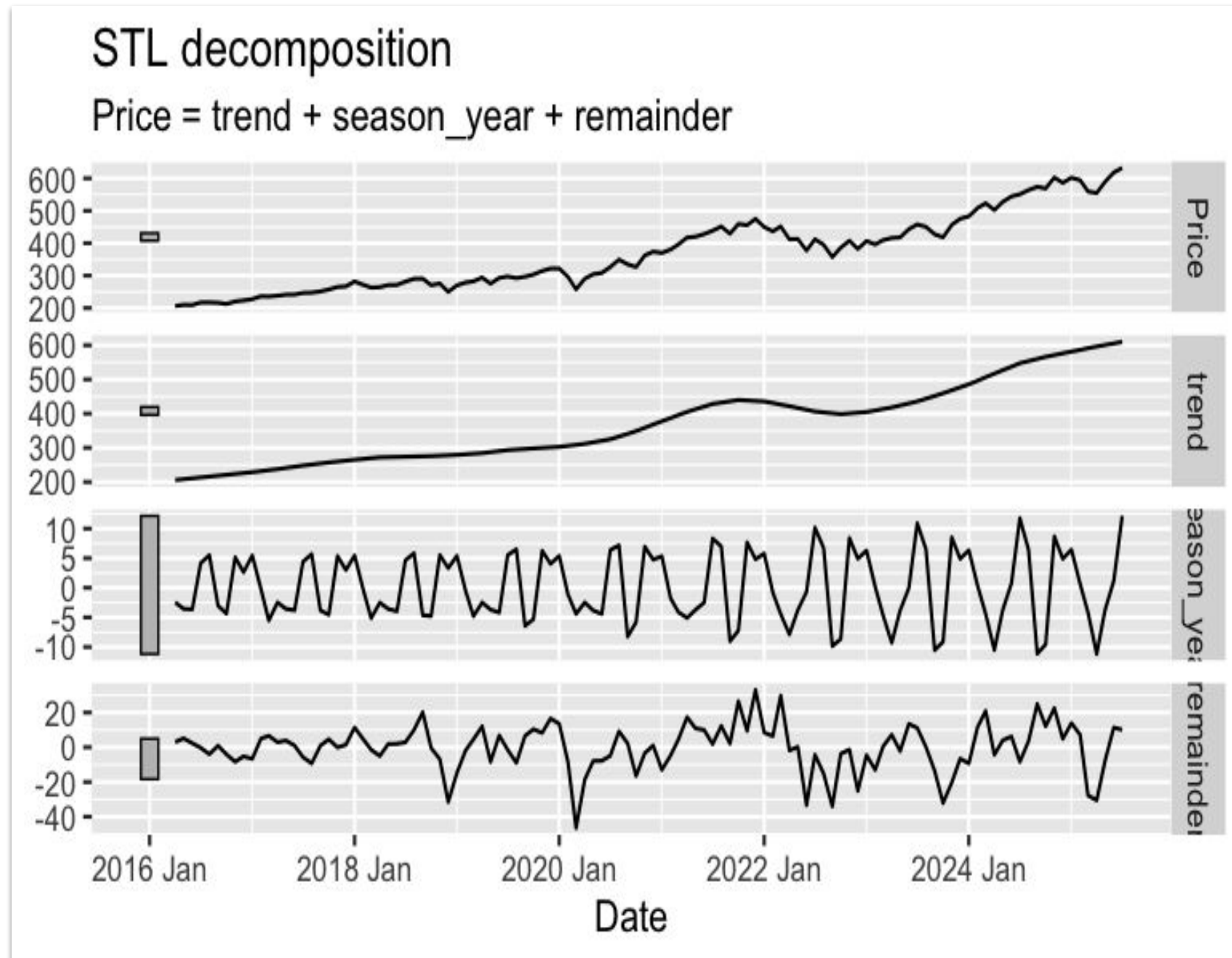
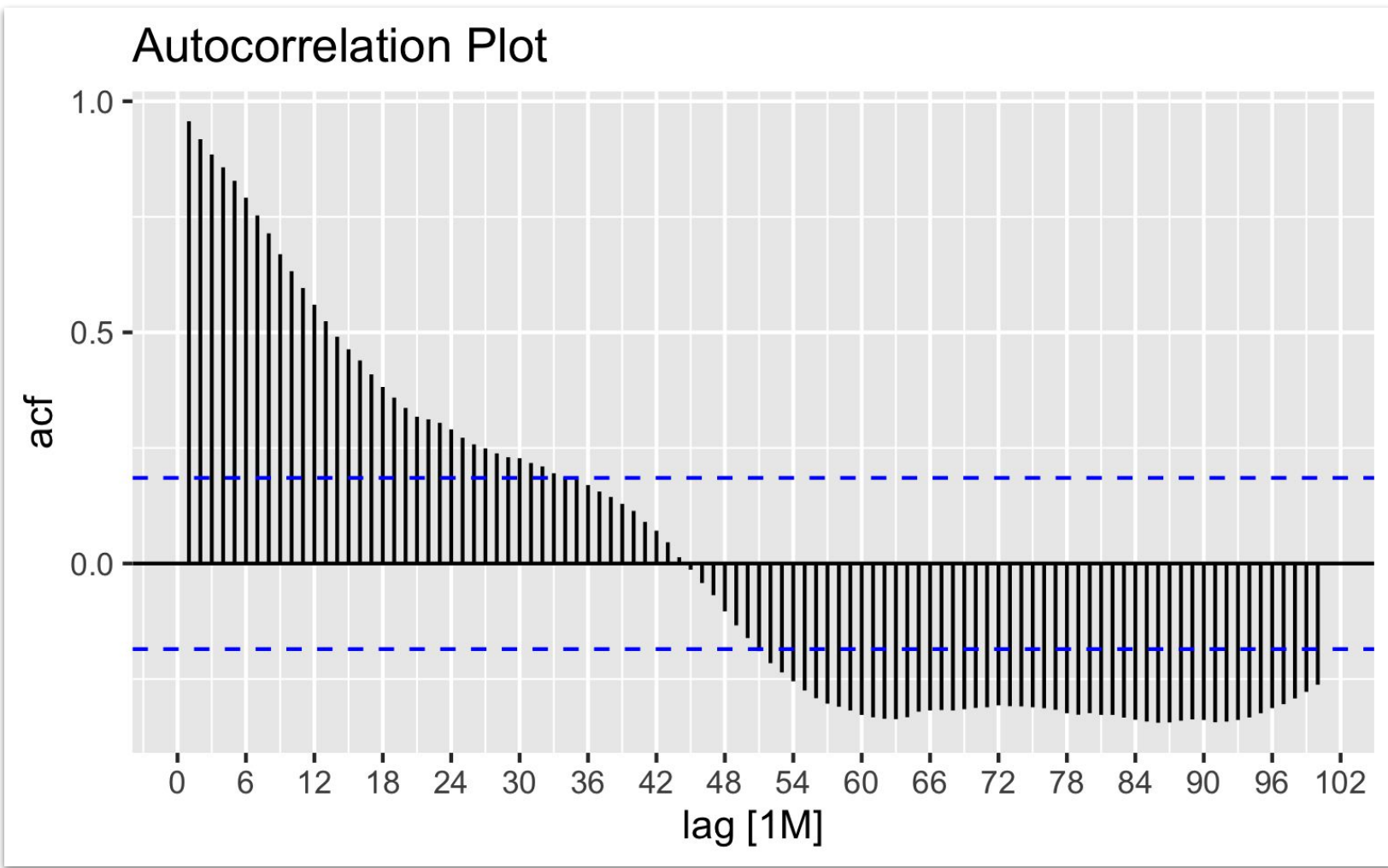
There are some theories on how to predict market performance. Yates (2024) discusses momentum theory. Rooted in behavioral finance, it assumes that market movements will continue in the same direction, but only in the short-term. Over longer periods, this effects reverses. Pande and Kumar (2025) acknowledged traditional methods to forecast stock prices and indices such as stochastic time series models (e.g., ARIMA, jump diffusion) but also modern machine learning methods (e.g., FFN, LSTM). The study found that LSTM models performed best for forecasting over almost all indices over all time horizons except for the S&P 500 and Nikkei 225 10 days ahead, where ARFIMA performed best.

Shih et al. (2024) wrote a journal article comparing deep learning models to more traditional models, such as Fama-French 3-factor models. This study found similar success in LSTM models and used them in combinations with the Fama-French models. Ngwaba (2025) takes a unique approach, focusing on forecasting applications for covered call ETFs and analyzing machine and deep learning models. Here, RNNs outperformed other models with the lowest MAE.

Limitations in existing literature recognize that deep learning models require significant computation demands, extensive data needs, and fail to capture many key factors that influence stock and ETF prices, like macroeconomic and geopolitical events.

Methods

During data exploration and preparation, the raw dataset was reduced to only the temporal variable (month and year) and the closing price, mutating the temporal variable to a suitable monthly format and identifying any missing values. Data exploration also involved visualizing the data to understand any patterns like trends or seasonality through STL decomposition. Inflation-adjusted ETF price values (using Core CPI) were also tested, however, it did not change behavioral data patterns. Data visualization and decomposition provided context into which models to experiment with and why certain models might forecast SPY-USA prices better than others in the results. The data was split 80/20 into training and testing data, respectively.

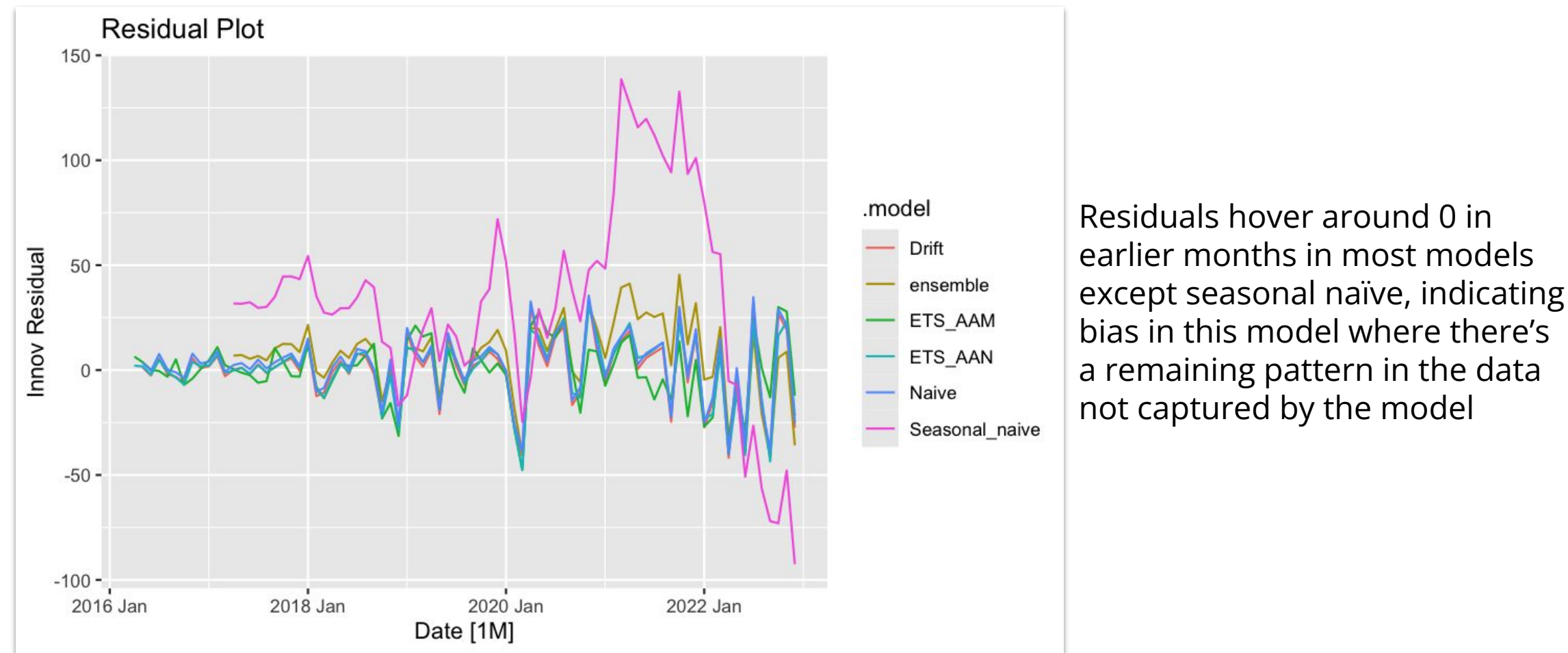
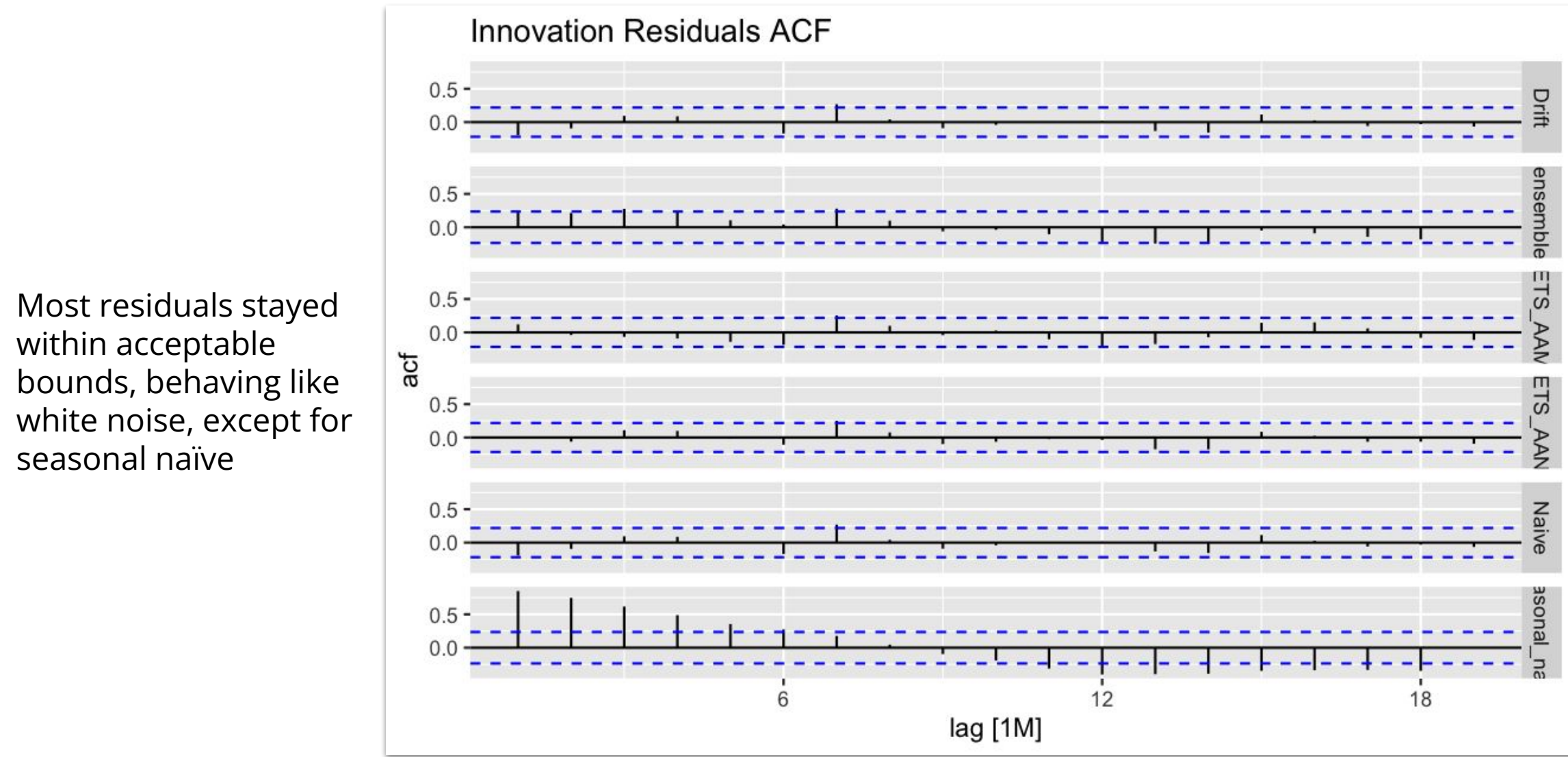
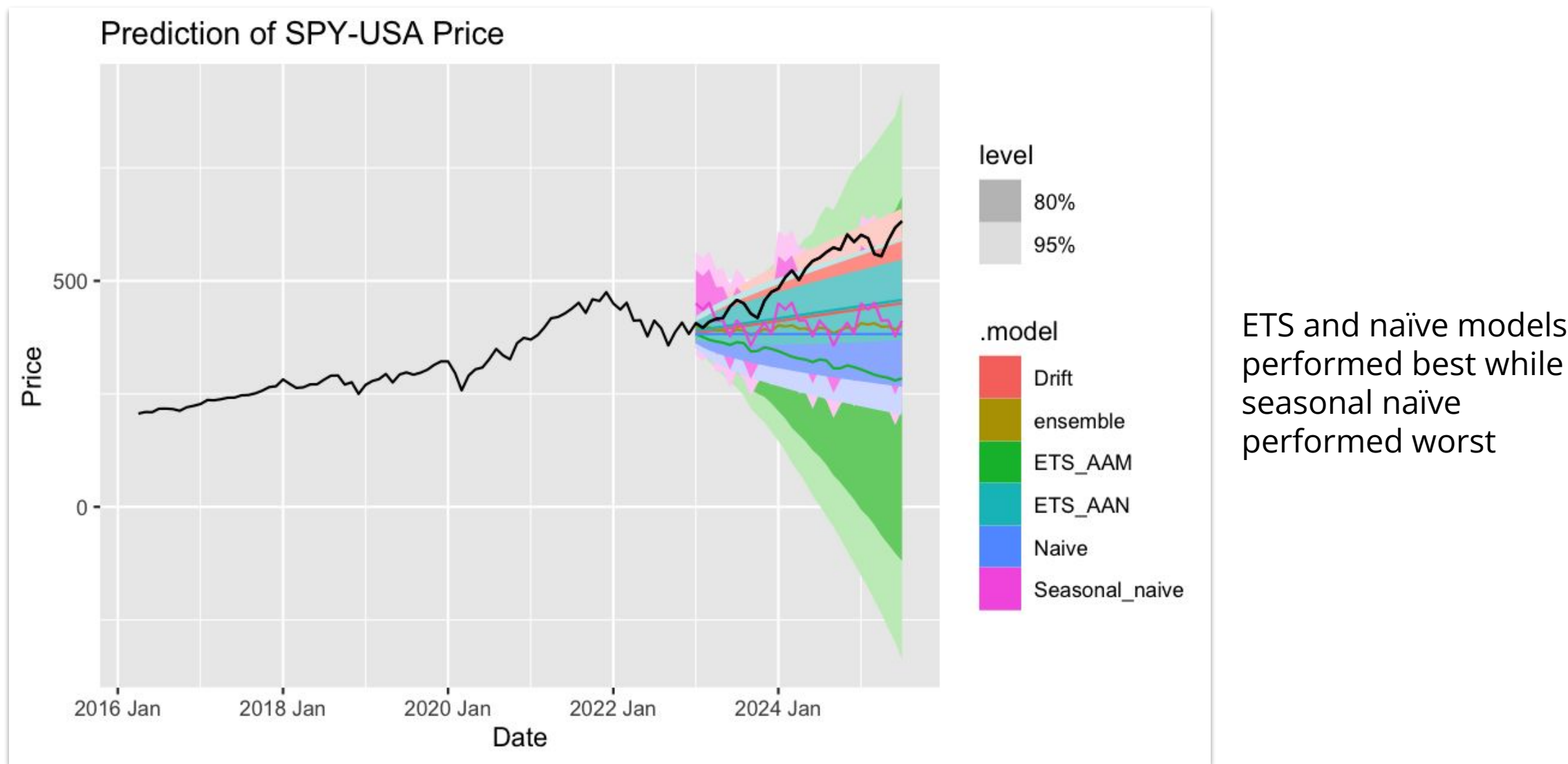


Data shows a smooth upward trend and artificial seasonality with increasing magnitude (multiplicative effect)

Results

Several models were tested and compared to determine which is the best for predicting the SPY-USA: ETS with automated parameter selection (with extensions in Holt Linear Trend and Holt-Winters), drift, naïve, seasonal naïve, and an ensemble model that combined forecasts using averaging.

Overall, the ETS_AAN (Holt's Linear Trend) and drift models performed best, achieving MAPE of 3.46% and 3.55% respectively. Its success is due to the fact that ETS_AAN can handle non-stationary data well and models an additive trend and error present in the data, while excluding seasonality. Trailing these models in performance was the naïve, ensemble, and seasonal naïve models. The ensemble model's performance was worsened by the seasonal naïve model which failed to capture the true pattern in the data, instead forcing a false seasonal pattern and overfitting for noise. Residual checks were conducted to verify these results and check statistical assumptions.



Discussion

The forecasting process involves first considering data patterns and then experimenting with various models and validating results to decide which most closely model the data's behavior.

This study considers a popular financial investment, the SPY-USA ETF, that tracks the S&P 500 and the different models that can effectively forecast performance. This study focused on low computational forecasting methods for tech savvy investors. It also explores an ensemble model, something mentioned as an area of further exploration in existing literature. The results were expected given the characteristics of financial data. ETS models have flexibility to adapt to different components like upward trend and irregular noise, while a drift model matches the “random walk” behavior often exhibited in ETF prices.

Some limitations of this study include not considering more advanced machine or deep learning models that other studies analyzed. While it does explore ensemble methods that average results from many models, it doesn't include hybrid models that may have produced better results, like ETS_AAN and LSTM combined. Another limitation is it doesn't consider other factors that could influence ETF performance that are challenging to model, such as geopolitical events.

Investors should consider simple and moderately complex models like drift, naïve and ETS as information to incorporate into investment decisions, like investing in the S&P 500. However, investors should not rely solely on these forecasts and should incorporate judgmental forecasting and domain knowledge to improve results.

Conclusion

Simple models are highly effective for financial forecasting. The additive ETS model and drift model best captured SPY-USA price behavior, outperforming seasonal approaches. Results emphasize the importance of aligning model structure with data characteristics, particularly avoiding overfitting noise as seasonality. Limitations of this work include its focus on univariate historical prices and omission of exogenous factors as well as more advanced machine and deep learning approaches.

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